

Autonomous Driving in Smart Cities: Deep Learning for Traffic Optimization

Deep Learning-Based Vehicle Trajectory Analysis for Highway Traffic Optimization

1st Moye Nyuysoni Glein Perry
moye-nyuysoni.glein-perry@stud.hshl.de
Deep Learning
Electronic Engineering
Hochschule Hamm Lippstadt

Abstract—Traffic congestion and unstable driving behavior remain major challenges in modern smart city transportation systems. Recent advances in deep learning offer promising opportunities to improve traffic management through data-driven prediction and optimization. In this study, a deep learning-based traffic optimization framework is proposed using naturalistic driving data from the HighD dataset. Raw vehicle trajectories are aggregated into lane-segment traffic states and used to train a multi-task Long Short-Term Memory (LSTM) model that predicts short-term future traffic speed and lane-change activity. Lane-change activity is treated as a proxy for traffic instability, enabling the system to capture dynamic traffic conditions beyond speed alone. Instead of directly predicting rare braking events, a rule-based variable speed limit strategy is employed to translate model predictions into proactive traffic control actions. A closed-loop evaluation is conducted using an acceleration-based braking proxy to assess the impact of the proposed approach. Experimental results demonstrate accurate traffic speed prediction, reliable detection of lane-change activity, and improved traffic stability under optimized speed profiles, with reduced speed variance and fewer harsh braking indicators compared to baseline behavior. The findings highlight the effectiveness of combining deep learning-based traffic prediction with interpretable optimization strategies for smart city traffic management.

KEYWORDS

Deep Reinforcement Learning (DRL), Intelligent Transportation Systems (ITS), Smart Cities (SC), Traffic Optimization (TO), Reinforcement Learning (RL), Deep Q-Network (DQN), Proximal Policy Optimization (PPO), State-Action-Reward (SAR), Vehicle Queue Length (VQL), Traffic Flow Management (TFM), Real-Time Signal Control (RTSC), Multi-Agent Systems (MAS), Urban Traffic Congestion (UTC), Dynamic Signal Timing (DST), Adaptive Traffic Signal Control (ATSC)

I. INTRODUCTION

The rapid advancement of compact, self-driving systems has transformed urban environments, making smart vehicles a practical choice for everyday travel. These autonomous systems reduce the demands placed on human drivers by carrying out tasks such as avoiding collisions, detecting lane departures, and recognizing traffic signs. Moreover, autonomous driving can optimize traffic movement, ease congestion, and improve

fuel efficiency by lowering emissions. Such vehicles also improve daily life by offering reliable and safe mobility for older adults and individuals with disabilities, reducing parking challenges, and preventing many accidents that traditionally stem from human error. Figure 1 presents an overview of how autonomous vehicles operate within smart cities. [7]



Fig. 1. An illustration of autonomous driving cars in smart cities [8]

As cities become smarter and more connected, technology is reshaping the way urban systems operate, including how traffic is managed. With more vehicles on the road than ever before, the need for intelligent traffic control has become a key priority for smart city development. Adaptive Traffic Signal Control (ATSC) is one of the most promising solutions. It adjusts traffic light timings in real time based on actual traffic flow, helping reduce congestion, improve travel times, and cut down on emissions. [6]

However, many of today's adaptive systems still rely on fixed rules or simplified traffic models. These approaches often fail in real world conditions, especially when traffic patterns change unexpectedly due to accidents, weather, or seasonal events. As a result, traditional systems struggle to efficiently manage traffic across large, complex networks of intersections, highlighting the need for more flexible, data-driven solutions.

While much of the existing research in smart city traffic management focuses on signalized urban intersections, highways are equally critical components of the urban transport network. Congestion, accidents, and unstable traffic flow on highways can spread to city streets and significantly reduce

overall mobility. Therefore, there is a strong motivation to apply advanced, data-driven methods, such as deep learning, not only to intersection control but also to highway traffic management, for example, through intelligent variable speed limits, ramp metering, and support for autonomous vehicles.

Deep reinforcement learning is a promising approach to adaptively adjust traffic light control policies based on real-time traffic situation, for example, traffic performance on each road [5]. Reinforcement learning, particularly in the form of Deep Reinforcement Learning, has emerged as a powerful approach for building truly adaptive traffic signal control systems. Unlike traditional methods that depend on fixed rules or predefined models, DRL allows the system to learn directly from real or simulated traffic environments. Through continuous interaction and feedback, the system develops strategies that improve over time based on actual traffic outcomes. By leveraging DRL, traffic signals can make intelligent, real-time adjustments to their timing and coordination, leading to smoother traffic flow, reduced congestion, and improved overall transportation efficiency. [6]

This paper presents a deep learning based framework for highway traffic optimization within smart city transportation systems. Unlike much of the existing literature, which primarily focuses on urban intersections and Adaptive Traffic Signal Control, this study targets highway environments. Highways play a critical role in regional mobility and are often major sources of large-scale congestion, speed instability, and unsafe driving behavior. Despite their importance, they remain relatively underexplored in data-driven traffic optimization research.

Using the HighD dataset, a high-resolution naturalistic highway trajectory dataset collected by drone, this work develops a predictive traffic modeling framework that combines deep learning-based traffic state forecasting with interpretable, rule-based traffic control. Raw vehicle trajectories are aggregated into lane-segment traffic states, enabling traffic-level analysis suitable for control-oriented applications.

The deep learning component is designed to predict short-term future highway traffic behavior, with a particular focus on mean traffic speed and lane-change activity. Speed is treated as the primary regression target, while lane-change activity serves as a proxy for traffic instability and turbulence. This multi-task learning formulation allows the model to capture interactions between speed dynamics and lateral driving behavior without directly predicting rare braking events, which are highly imbalanced in naturalistic driving data.

Rather than relying on reinforcement learning, traffic optimization is achieved through a rule-based variable speed limit strategy that translates model predictions into proactive control actions. Predicted speed and lane-change risk are used to generate speed advisories aimed at improving traffic stability and reducing abrupt speed variations. The effectiveness of the proposed approach is evaluated using a closed-loop proxy, in which recommended speed limits are applied to observed traffic speed profiles and braking-related indicators are derived from an acceleration-based proxy.

Through this integrated prediction-and-control framework, the proposed approach demonstrates the potential of deep learning-based traffic state estimation to support practical and interpretable highway traffic optimization in smart city contexts.

The main contributions of this paper are as follows:

- We propose a deep learning-based framework for highway traffic optimization using naturalistic driving data from the HighD dataset.
- We develop a multi-task LSTM model that jointly predicts short-term traffic speed and lane-change activity at the lane-segment level.
- We introduce a rule-based variable speed limit strategy that leverages model predictions to improve traffic stability without relying on reinforcement learning.
- We evaluate the proposed framework using a closed-loop proxy and demonstrate reductions in speed variance and braking-related indicators compared to baseline traffic behavior.

The remainder of this paper is structured as follows: Section II gives a system overview using an activity, III provides a comprehensive background on deep and reinforcement learning and existing traffic signal control methodologies. Section III-C describes the HighD dataset and the traffic environment considered in this study. Section IV outlines the proposed architecture and implementation details. Section V presents the experimental setup and performance results. Finally, Section VI concludes the paper and suggests directions for future research.

II. SYSTEM DESIGN OVERVIEW

A. Activity Diagram

The activity diagram illustrates the workflow of the proposed deep learning based traffic optimization framework using the HighD dataset. Vehicle trajectories are first loaded and aggregated into segmented lane traffic state sequences, which are then split into training and validation sets. The system either trains a multi-task LSTM model or loads a previously trained model. The trained model is then used in a closed-loop evaluation to predict short-term traffic speed and lane-change risk. These predictions are provided to a rule-based variable speed limit module to compute speed advisories to improve traffic stability. Finally, performance metrics and visualizations are generated to assess prediction accuracy and braking-related improvements

III. BACKGROUND AND LITERATURE REVIEW

In smart cities, where efficient mobility depends on accurate traffic prediction and adaptive control strategies, traffic optimization is a key part of autonomous driving systems. Deep learning has made it possible to create more advanced models for analyzing large amounts of traffic data. These models can be used for things like predicting highway flow, modeling lane-change behavior, and coordinating smart infrastructure.

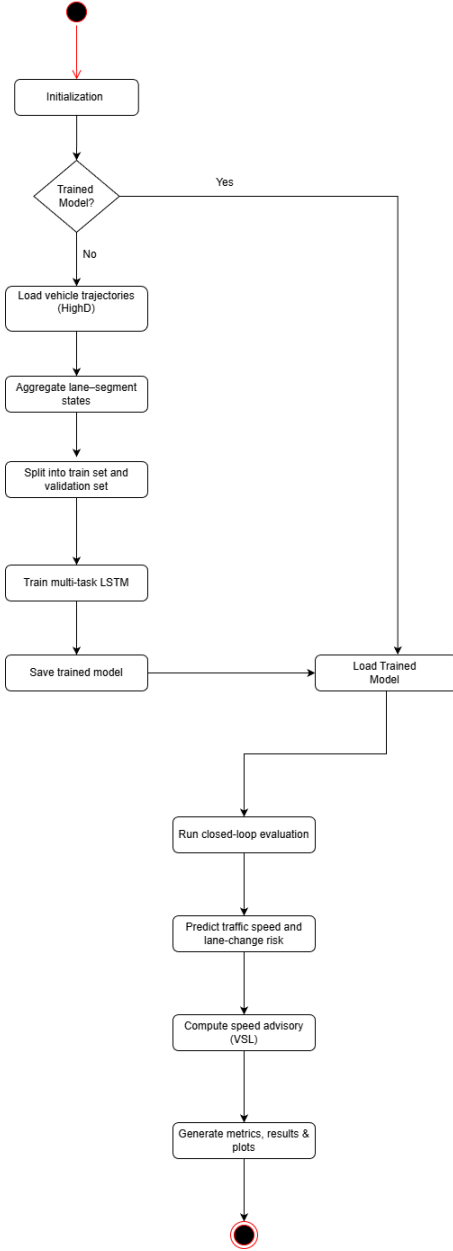


Fig. 2. Activity Diagram [2]

A. Deep Learning Approaches for Traffic Optimization

Recent studies have explored the use of deep reinforcement learning (DRL) to improve traffic control efficiency. It was demonstrated that deep reinforcement learning-based adaptive traffic signal control can significantly decrease vehicle delay and enhance traffic throughput in urban settings. The study focuses on signal optimization at specific intersections, yet it highlights the broader relevance of DRL methodologies for dynamic and scalable traffic management within smart city infrastructures. [6]

Recent work has also shown that DRL can generate more coordinated and efficient traffic signal behaviors across larger networks. In a study examining both single intersections

and grid-based road networks, the authors found that DRL algorithms such as DQN and DDPG can naturally learn optimal control strategies, including the emergence of the “greenwave” pattern that enables continuous vehicle flow. Their results demonstrate not only the ability of DRL to match optimal policies at isolated intersections but also its scalability in producing intelligent, network-wide coordination without explicit programming. [5]

CityFlowER proposes an embedded learning-based traffic simulator in which neural network models are trained to imitate vehicle behaviors such as speed adjustment and lane-changing decisions. These models are pre-trained from trajectory data and directly integrated into the simulator, replacing traditional rule-based car-following and lane-changing models. Experimental results demonstrate that CityFlowER can accurately reproduce realistic driving behaviors while maintaining high computational efficiency, even in large-scale traffic scenarios. This work highlights the effectiveness of data-driven traffic behavior prediction for improving traffic modeling and supports the use of learned speed and lane-change predictions as a foundation for traffic optimization strategies. [4]

B. Introduction to Deep Learning

Deep learning is a supervised learning paradigm that extends classical neural networks by employing multiple hidden layers to learn increasingly abstract representations of data. [3] deep learning models are capable of automatically extracting meaningful features from input data through hierarchical transformations, thereby reducing the need for manual feature engineering that is common in traditional machine learning approaches.

A deep neural network is composed of several layers of artificial neurons, where each neuron computes a weighted sum of its inputs and applies a non-linear activation function. Formally, the output of a neuron can be written as:

$$y = f(\mathbf{w} \cdot \mathbf{x} + b), \quad (1)$$

where $\mathbf{x}$ denotes the input vector, $\mathbf{w}$ represents the trainable weight parameters, b is the bias term, and $f(\cdot)$ is a non-linear activation function. The introduction of non-linearity enables deep networks to approximate complex functional relationships between inputs and outputs.

Training deep neural networks involves adjusting the weights and biases to minimize a predefined loss function that measures the discrepancy between predicted and target outputs. This optimization process is performed using backpropagation, where gradients of the loss function are propagated backward through the network layers and parameters are updated iteratively using gradient-based optimization techniques [3]. Through this process, the network progressively refines its internal representations.

In traffic modeling applications, deep learning is particularly effective due to its ability to process large volumes of data and capture non-linear temporal patterns. Highway traffic behavior is influenced by historical speed variations,

vehicle interactions, and changing traffic density, all of which evolve over time. Deep learning models are well suited to learn such dependencies and provide accurate short-term traffic predictions.

In this work, deep learning is applied to model future highway traffic states using historical traffic sequences extracted from the HighD dataset. The learned representations are used to predict traffic speed and lane-change activity, which serve as key inputs for the proposed traffic optimization framework.

C. Recurrent Neural Networks and Long Short-Term Memory

Recurrent Neural Networks are a class of neural networks specifically designed for sequential and time-dependent data. Unlike feedforward neural networks, RNNs incorporate feedback connections that allow information from previous time steps to influence the current output [1], this recurrent structure enables RNNs to model temporal dependencies by maintaining an internal state that evolves as new inputs are processed. Such a capability makes RNNs suitable for applications involving sequences, including speech processing, natural language modeling, and time-series prediction.

Despite their strengths, standard RNNs suffer from significant limitations when learning long-term dependencies. During training, gradients propagated through many time steps can either vanish or explode, making it difficult for the network to retain information from distant past inputs. This issue restricts the effectiveness of basic RNNs in scenarios where long-range temporal relationships are important.

Long Short-Term Memory (LSTM) networks were introduced to address these limitations of conventional RNNs. LSTMs enhance the recurrent architecture by incorporating an explicit memory mechanism that allows the network to selectively store, update, and retrieve information over extended time horizons [1], this is achieved through gating structures that regulate information flow, enabling the network to preserve relevant temporal information while discarding irrelevant data.

Because of their ability to capture both short-term variations and long-term trends, LSTM networks are particularly well suited for modeling complex sequential processes. In traffic modeling applications, LSTMs can effectively learn how traffic states evolve over time by considering historical speed patterns, traffic density changes, and interactions between vehicles. This makes them a natural choice for predicting future traffic behavior in highway environments.

In this study, an LSTM-based architecture is employed to model sequences of aggregated highway traffic states. By leveraging the temporal modeling capabilities of LSTMs, the proposed framework is able to generate reliable short-term predictions of traffic speed and lane-change activity, which are subsequently used for traffic optimization.

sectionDataset and Traffic Environment

D. HighD Dataset Description

To ensure realism and practical relevance in highway traffic modeling, this study makes use of the HighD dataset, a



Fig. 3. Illustration of the HighD dataset and recording setup [8]

large scale naturalistic driving dataset specifically designed for highway environments. Unlike many traffic datasets that focus on urban intersections or rely on simulation, HighD captures real-world highway traffic behavior with high spatial and temporal accuracy [8].

The dataset was collected using drones equipped with high-resolution cameras, recording traffic from an aerial perspective on multiple German highway segments. Vehicle trajectories were extracted using advanced computer vision techniques, resulting in precise microscopic measurements of vehicle motion. The dataset includes detailed information for many of vehicles, such as lane assignments, longitudinal and lateral positions, velocities, accelerations, vehicle dimensions, and time headways.

Compared to earlier highway datasets such as NGSIM, HighD offers a larger scale and greater diversity, with recordings captured at different locations and times of day. This diversity leads to a wide range of traffic conditions, including free flow, transitional, and congested states. Reported speed distributions show clear peaks around typical highway driving speeds, making the dataset well suited for learning-based traffic modeling and prediction tasks [8].

A key advantage of the HighD dataset is its high temporal resolution, with trajectories typically recorded at 25 Hz. This allows for fine-grained analysis of traffic dynamics and enables reliable aggregation into traffic-state representations at coarser temporal scales.

Since HighD focuses exclusively on multi-lane highways without signalized intersections, it is particularly appropriate for highway traffic optimization tasks such as speed prediction, traffic stability analysis, and control strategies relevant to connected and autonomous vehicles.

E. Data Loading and Preprocessing

Traffic data were loaded using a custom data loading pipeline that processes the three core files provided for each HighD recording. The `tracks.csv` file contains frame-level vehicle trajectory data, `tracksMeta.csv` provides vehicle-level metadata, and `recordingMeta.csv` includes recording-level information such as lane configuration, frame rate, and roadway geometry.

Each recording was processed independently to preserve temporal consistency. Raw trajectory data were first sorted by vehicle identifier and frame index to ensure correct temporal ordering. Vehicle-level trajectories were then aggregated into lane-segment traffic states using fixed temporal bins. This aggregation step converts microscopic vehicle data into traffic-level representations that are more suitable for prediction and control applications.

For each lane segment and time interval, traffic-state variables such as mean speed, traffic density, flow, and lane-change activity were computed. These aggregated traffic states form multivariate time-series sequences that serve as inputs to the deep learning model. Feature normalization was applied using statistics computed from the training data to ensure stable and consistent learning behavior.

IV. METHODOLOGY

A. System Architecture and Overall Framework

This work proposes a data-driven traffic optimization framework built on naturalistic driving data from the HighD dataset. The system follows a sequential architecture that transforms raw vehicle trajectories into traffic-level predictions and optimization decisions. Rather than operating at the individual vehicle level, the framework focuses on aggregated traffic states at the lane-segment level, which is more relevant for traffic management in smart city contexts.

The architecture consists of four main stages: data loading and preprocessing, traffic-state aggregation, traffic prediction using deep learning, and traffic optimization through speed advisory generation. Raw trajectories are first extracted from multiple HighD recordings and converted into structured traffic states. These states are then used to train a predictive model that estimates short-term traffic behavior. Finally, model predictions are translated into speed control decisions and evaluated using a closed-loop proxy.

B. Traffic State Representation and Feature Extraction

The HighD dataset provides detailed vehicle trajectories recorded at high temporal resolution. To reduce noise and allow traffic-level analysis, vehicle trajectories are aggregated into fixed lane segment units over uniform time intervals. Each segment of lane at a given time step is represented by a set of traffic-state variables that capture both traffic flow and stability.

The main features extracted for each segment of the lane include the mean speed, which represents the average velocity of traffic. Traffic density computed as the number of vehicles per segment, Traffic flow, derived from speed and density, Lane-change activity expressed as a binary or rate-based indicator of lane-change occurrence.

These features collectively describe the dynamic state of the traffic and serve as input to the predictive model. The aggregated traffic states are organized as time series data, allowing the model to learn temporal patterns and dependencies in the evolution of the traffic.

C. Deep Learning-Based Traffic Prediction Model

To predict short-term traffic behavior, a multi-task Long Short-Term Memory (LSTM) neural network is employed. LSTM networks are particularly well-suited for sequential data and are capable of capturing temporal dependencies in traffic dynamics. The model uses a shared recurrent structure followed by two task-specific output heads.

The first output head performs regression to predict the future mean speed of traffic over a short prediction horizon.

The second output head performs binary classification to estimate lane-change activity, which acts as an indicator of traffic turbulence and instability. By learning both tasks simultaneously, the model captures interactions between speed variation and lane-change behavior, leading to more informative traffic predictions.

D. Training Strategy and Data Splitting

The aggregated traffic-state dataset is divided into training and validation subsets using a fixed split. Specifically, 80 percent of the data is used for training, while the remaining 20 percent is reserved for validation. The split is performed at the sequence level to prevent information leakage across time.

Before training, all input features are normalized using statistics computed exclusively on the training set. The model is trained in a supervised manner using a combined loss function. Speed prediction is optimized using a Huber loss to ensure robustness to outliers, while lane change prediction is optimized using a binary cross-entropy loss. The total loss is computed as a weighted sum of both objectives.

Training is carried out using the AdamW optimizer with early stopping based on validation performance. This prevents overfitting and ensures stable convergence.

E. Traffic Optimization via Speed Advisory Generation

Rather than directly predicting braking events, which are rare and highly imbalanced in naturalistic driving data, traffic optimization is addressed indirectly through speed regulation. A rule-based variable speed limit (VSL) optimizer is used to translate traffic predictions into actionable speed advisories.

The optimizer takes as input the predicted future speed and lane-change risk and selects an appropriate speed limit from a predefined set of admissible values. Higher lane-change risk or excessive predicted speed leads to more conservative speed recommendations. Additional constraints are applied to prevent abrupt changes in speed limits, ensuring realistic and safe control behavior.

F. Evaluation and Closed-Loop Analysis

Model performance is evaluated at two levels: prediction accuracy and traffic-level impact. Prediction quality is assessed using standard regression and classification metrics, including mean absolute error and root mean squared error for speed prediction, as well as precision, recall, F1-score, and average precision for lane-change activity detection.

To assess the effectiveness of the proposed traffic optimization strategy, a closed-loop evaluation proxy is employed. In this setup, recommended speed limits are applied to observed traffic speed profiles, and the resulting acceleration patterns are analyzed. Braking-related indicators derived from an acceleration proxy are compared between baseline and optimized scenarios to evaluate improvements in traffic smoothness and stability.

G. Implementation Details

The proposed framework is implemented in Python using the PyTorch deep learning library. Data preprocessing and feature aggregation are modularized to support scalability across multiple HighD recordings. Visualization tools are used to analyze learning curves, prediction results, and the impact of speed advisories on traffic behavior.

V. RESULTS

A. Traffic Speed Prediction Performance

The proposed multi-task LSTM model demonstrates strong performance in short-term traffic speed prediction. Across the validation set, the model achieves a mean absolute error (MAE) of approximately 1.0 m/s and a root mean squared error (RMSE) between 1.7 and 1.9 m/s, indicating accurate estimation of future traffic speed at the lane-segment level.

Training and validation curves show stable convergence, with no significant divergence between training loss and validation error. This suggests that the model generalizes well and does not suffer from severe overfitting. The use of a Huber loss for speed regression contributes to robustness against occasional speed outliers present in naturalistic traffic data.

A comparison between actual and predicted speed values further confirms the quality of the predictions. The predicted speeds closely follow the ground truth across a wide range of traffic conditions, including both moderate and high-speed regimes.

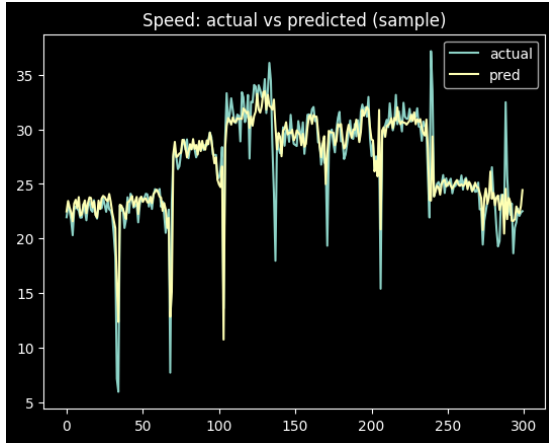


Fig. 4. An illustration of Actual Vs Predicted Speeds [2]

B. Lane-Change Activity Prediction Results

In addition to speed prediction, the model effectively identifies lane-change activity as a proxy for traffic instability. On the validation set, the lane-change prediction task achieves precision values around 0.86–0.87, recall values around 0.90–0.92, and an F1-score close to 0.89. The average precision (PR-AUC) remains consistently high at approximately 0.92, indicating strong discriminative capability despite class imbalance.

The confusion matrix shows that the model correctly identifies most lane-change events while maintaining a reasonable false-positive rate. This balance between precision and recall is particularly important in traffic optimization, where missing instability events can be more detrimental than occasional conservative predictions.

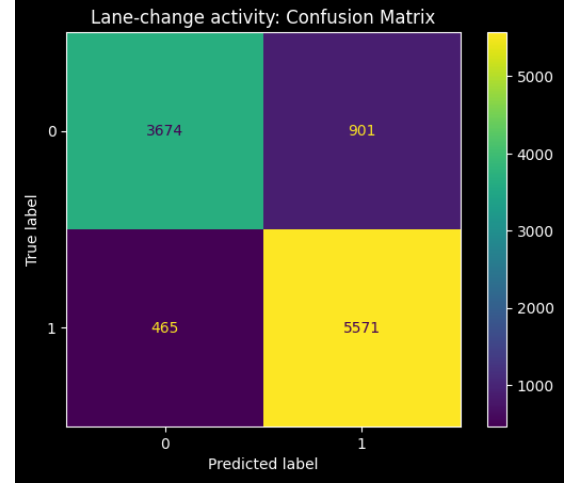


Fig. 5. An illustration of Confusion Matrix of lane change [2]

C. Training Stability and Multi-Task Learning Behavior

Analysis of the training loss components reveals that both tasks contribute meaningfully to the learning process. The total loss decreases rapidly during early epochs and stabilizes after several iterations. The speed regression loss dominates the total loss magnitude, while the lane-change classification loss remains stable and well-controlled.

Early stopping based on validation MAE prevents over-training and ensures that the selected model corresponds to the best generalization performance. Overall, the learning dynamics indicate that joint training of speed prediction and lane-change activity is stable and effective.

D. Closed-Loop Traffic Optimization Results

To evaluate the impact of traffic prediction on optimization, a closed-loop evaluation proxy is employed. In this setup, predicted traffic speed and lane-change risk are passed to a rule-based variable speed limit (VSL) optimizer, which generates speed advisories that are applied to observed traffic speed profiles.

Results show a clear improvement in traffic smoothness under the optimized scenario. The optimized speed profile exhibits lower variance compared to the baseline, indicating reduced speed fluctuations. Furthermore, braking-related indicators derived from an acceleration proxy show a reduction in harsh deceleration events and overall braking severity.

The acceleration proxy distribution becomes more concentrated around zero after optimization, suggesting smoother longitudinal behavior. These results indicate that the proposed

framework can effectively reduce traffic instability by proactively regulating speed based on predicted traffic conditions.



Fig. 6. An illustration of Closed Loop Predictions [2]

E. Discussion of Results

The results confirm that indirect traffic optimization through speed prediction and lane-change awareness is more effective and robust than direct prediction of rare braking events. By focusing on smoother speed evolution and traffic stability, the proposed approach avoids issues related to class imbalance while still achieving meaningful reductions in braking-related indicators.

Overall, the experimental findings demonstrate that combining deep learning-based traffic prediction with a rule-based optimization strategy can improve traffic flow stability and contribute to safer and more efficient traffic management in smart city environments.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This study presented a deep learning-based framework for traffic optimization in smart city environments using naturalistic driving data from the HighD dataset. By aggregating raw vehicle trajectories into lane-segment traffic states, the proposed system focuses on traffic-level behavior rather than individual vehicle actions. A multi-task LSTM model was employed to predict short-term future traffic speed and lane-change activity, capturing both flow evolution and traffic instability.

Experimental results demonstrate that the proposed model achieves accurate speed prediction and reliable detection of lane-change activity, with stable training behavior and strong generalization performance. Rather than directly predicting rare braking events, traffic optimization was addressed indirectly through a rule-based variable speed limit strategy. Closed-loop evaluation results show that this approach leads to smoother speed profiles, reduced speed variance, and a noticeable reduction in braking-related indicators derived from an acceleration proxy.

Overall, the findings confirm that combining deep learning-based traffic prediction with interpretable rule-based control can effectively improve traffic stability. The proposed framework provides a practical and scalable solution for traffic management, bridging the gap between data-driven prediction and actionable traffic optimization.

B. Future Work

Several directions can be explored to extend this work. First, the proposed closed-loop evaluation relies on a proxy-based simulation. Future studies could integrate the framework into a microscopic traffic simulator or real-time traffic management system to validate its effectiveness under more realistic control dynamics.

Second, additional traffic features such as weather conditions, road geometry, or multi-lane interactions could be incorporated to further improve prediction accuracy. Extending the model to multi-horizon forecasting would also allow for more flexible and adaptive traffic control strategies.

Finally, while this study adopts a rule-based optimization approach for interpretability and stability, future work could investigate reinforcement learning-based controllers that learn optimal speed control policies directly from traffic feedback. Hybrid approaches combining learned policies with safety constraints may offer a promising direction for advanced smart city traffic management systems.

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